

TASK 3

Task 3.1: Create a Dockerfile for your Python Flask "Hello World" application. The Dockerfile should build a minimal image that runs the application on port 5000. Build the Docker image locally and verify it runs correctly.

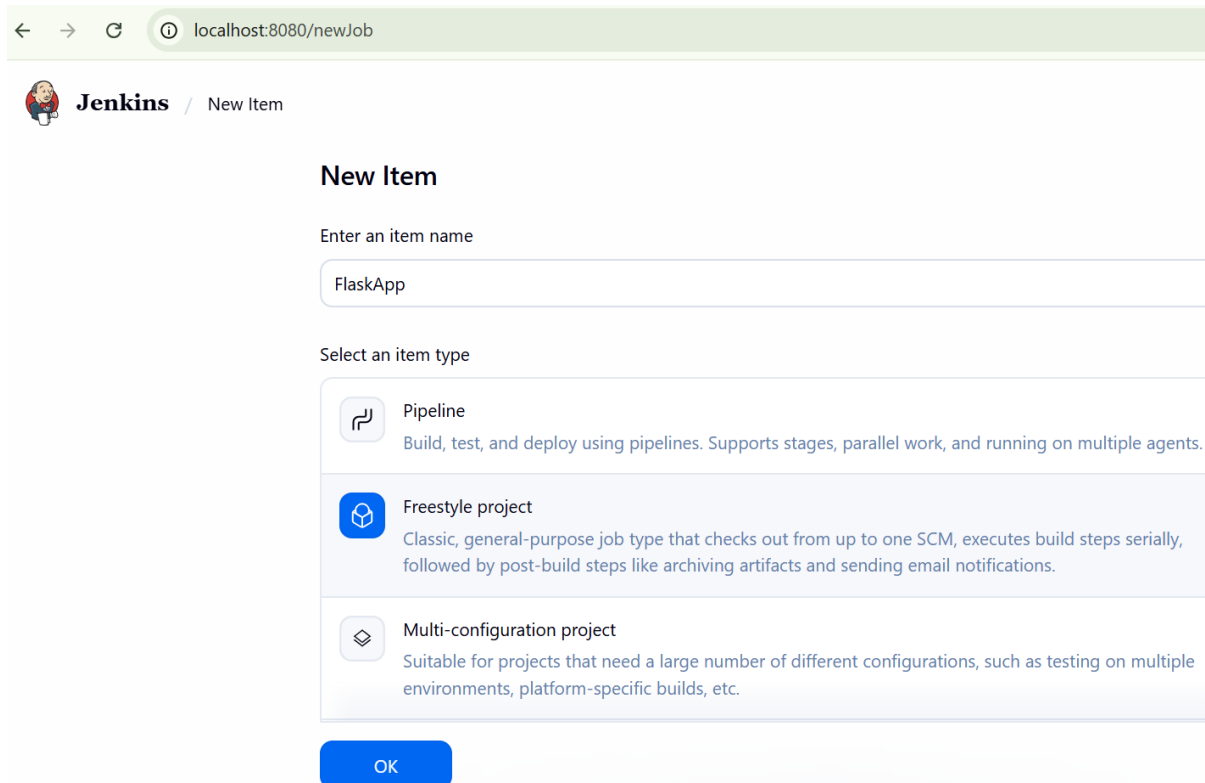
The screenshot shows the Visual Studio Code interface with the following components:

- EXPLORER:** Shows the file structure with 'Dockerfile' and 'app.py' under the 'FLASKAPP' folder.
- app.py:** Contains the following code:

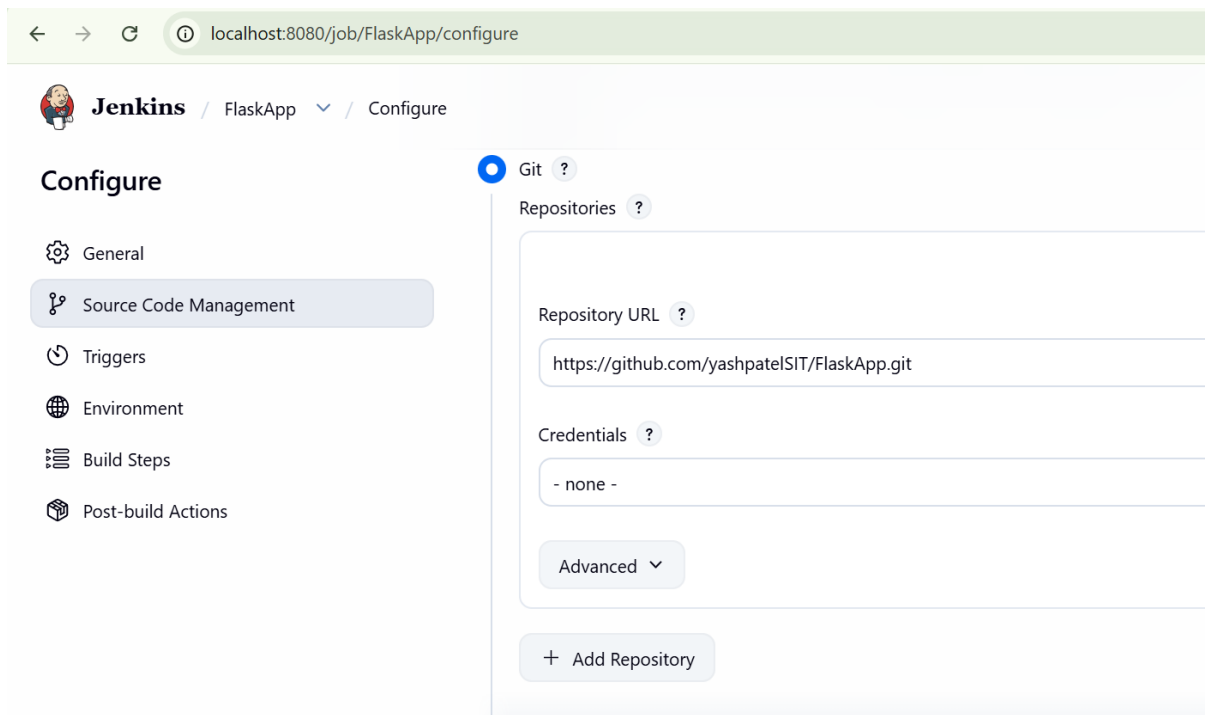
```
9 def home():
10     return "Hello World!"
11
12 if __name__ == '__main__':
13     app.run(host="0.0.0.0", port=5000, debug=True)
```
- TERMINAL:** Shows the command execution and output:

```
PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp> docker build -t flaskapp .
o PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp> docker run -p 5000:5000 flaskapp
Welcome to the Flask Application
User Authentication Feature
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 113-745-611
172.17.0.1 - - [26/Jul/2026 15:57:27] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [26/Jul/2026 15:57:27] "GET /favicon.ico HTTP/1.1" 404 -
```
- Browser:** Shows the URL '127.0.0.1:5000' with the response 'Hello World!'.

Task 3.2: Set up a Jenkins Freestyle project. Configure it to pull your Git repository (from Task 1.1) and execute a build step that lists the contents of the workspace. Trigger a build and provide a screenshot of the successful build output.



The screenshot shows the Jenkins 'New Item' page in a web browser. The address bar indicates the URL is localhost:8080/newJob. The page header shows the Jenkins logo and 'New Item'. The main section is titled 'New Item'. Below this, there is a text input field for 'Enter an item name' with the value 'FlaskApp'. Underneath, there is a section 'Select an item type' with three options: 'Pipeline' (described as 'Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.'), 'Freestyle project' (described as 'Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.'), and 'Multi-configuration project' (described as 'Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.'). The 'Freestyle project' option is selected. At the bottom of this section is a blue 'OK' button.



The screenshot shows the Jenkins 'Configure' page for the 'FlaskApp' job. The address bar indicates the URL is localhost:8080/job/FlaskApp/configure. The page header shows the Jenkins logo and 'FlaskApp / Configure'. The main section is titled 'Configure'. On the left, there is a sidebar with navigation links: 'General', 'Source Code Management' (selected), 'Triggers', 'Environment', 'Build Steps', and 'Post-build Actions'. The 'Source Code Management' section is expanded, showing a 'Git' repository type. Below this, there is a 'Repositories' section with a 'Repository URL' field containing 'https://github.com/yashpatelSIT/FlaskApp.git'. There is also a 'Credentials' field with a dropdown menu showing '- none -'. Below these fields is an 'Advanced' dropdown menu. At the bottom of the 'Repositories' section is a '+ Add Repository' button.

← → ↻ ⓘ localhost:8080/job/FlaskApp/configure

 **Jenkins** / FlaskApp ▾ / Configure

Configure

- ⚙️ General
- 🔑 Source Code Management
- 🕒 Triggers
- 🌐 Environment
- 📋 Build Steps
- 📦 Post-build Actions

Branch Specifier (blank for 'any') ?

*/main

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

+ Add

Triggers

← → ↻ ⓘ localhost:8080/job/FlaskApp/1/

 **Jenkins** / FlaskApp ▾ / #1

- 📄 Status
- 📄 Changes
- 📄 Console Output
- 🗑️ Delete build '#1'
- 📦 Git Build Data
- ✎️ Edit build information

✅ #1 (Jul 26, 2026, 9:34:47 PM)

🕒 Started by user [Yash Patel](#)

📦 git

Revision: 1d569738681588567fd23cc1aa678329896155d1
Repository: <https://github.com/yashpatelSIT/FlaskApp.git>

- refs/remotes/origin/main

📄 No changes.

localhost:8080/job/FlaskApp/1/console

Jenkins / FlaskApp / #1

Status
Changes
Console Output
Delete build '#1'
Git Build Data
Edit build information

Console

Download Copy View

```
Started by user Yash Patel
Running as SYSTEM
Building in workspace C:\Users\yashp\.jenkins\workspace\FlaskApp
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/yashpatelSIT/FlaskApp.git
> git.exe init C:\Users\yashp\.jenkins\workspace\FlaskApp # timeout=10
Fetching upstream changes from https://github.com/yashpatelSIT/FlaskApp.git
> git.exe --version # timeout=10
> git --version # 'git version 2.54.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/yashpatelSIT/FlaskApp.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe config remote.origin.url https://github.com/yashpatelSIT/FlaskApp.git # timeout=10
> git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 1d569738681588567fd23cc1aa678329896155d1 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 1d569738681588567fd23cc1aa678329896155d1 # timeout=10
Commit message: "Resolved merge conflict"
First time build. Skipping changelog.
[FlaskApp] $ cmd /c call C:\Users\yashp\AppData\Local\Temp\jenkins10429416318184405099.bat
C:\Users\yashp\.jenkins\workspace\FlaskApp\dir
```

localhost:8080/job/FlaskApp/1/console

Jenkins / FlaskApp / #1

```
Avoid second fetch
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 1d569738681588567fd23cc1aa678329896155d1 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 1d569738681588567fd23cc1aa678329896155d1 # timeout=10
Commit message: "Resolved merge conflict"
First time build. Skipping changelog.
[FlaskApp] $ cmd /c call C:\Users\yashp\AppData\Local\Temp\jenkins10429416318184405099.bat

C:\Users\yashp\.jenkins\workspace\FlaskApp>dir
Volume in drive C is Windows
Volume Serial Number is 0249-629A

Directory of C:\Users\yashp\.jenkins\workspace\FlaskApp

07/26/2026  09:34 PM    <DIR>          .
07/26/2026  09:34 PM    <DIR>          ..
07/26/2026  09:34 PM                245 app.py
               1 File(s)                245 bytes
               2 Dir(s)  229,543,272,448 bytes free

C:\Users\yashp\.jenkins\workspace\FlaskApp>exit 0
Finished: SUCCESS
```

The screenshot shows the Docker Desktop application window. The left sidebar contains navigation options: Gordon, Containers, Images (selected), Logs, Volumes, Kubernetes, Builds, Models, MCP Toolkit (BETA), Docker Hub, and Extensions. The main area is titled 'Images' and shows the 'Local' tab. It displays a table of images with the following data:

	Name	Tag	Image ID	Created	Size	Actions
☐	flaskapp	latest	2aec79eb43b3	36 minutes ag	213.25 MB	

Below the table, there are 'Walkthroughs' for 'How do I run a container?' (6 mins) and 'Run Docker Hub images' (5 mins). At the bottom, a status bar shows 'Engine running', RAM usage (0.96 GB), CPU usage (0.08%), and disk usage (1.70 GB used of 1006.85 GB limit).

The screenshot shows a GitHub repository page for 'yashpatelSIT / FlaskApp'. The repository is public and has 2 branches and 0 tags. The commit history shows a recent commit by 'yashpatelSIT' titled 'Add Dockerfile and finalize Flask application' with 7 commits. The file list includes 'Dockerfile' and 'app.py', both added in the same commit. The README section is currently empty, with a prompt to 'Add a README' to help people understand the project.